MATLAB R2022b

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- signalsandsystemexp18.m
- signalsandsystemexp15.m
- signalsandsystemexp13.m
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signalsandsystemexp10.m (Script)

Experiment 10: Zero-input, zero-state, impulse response and convolution

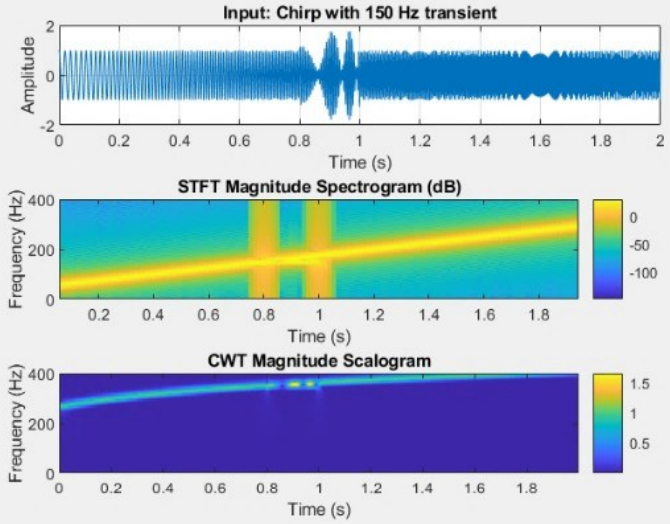
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signalsandsystem

```
16      subplot(
17      plot(t,x
18      xlabel('
19      title('I
20      subplot(
21      imagesc(
22      ylim([0
23      xlabel('
24      title('S
25      % Contin
26      subplot(
27      [wt,fCWT
28      imagesc(
29      ylim([0
30      xlabel('
31      title('C
32      % Report
33      fprintf(
34      fprintf('STFT window = %d samples; overlap = %d samples; FFT length = %d\n', ...
35      length(win),noverlap,nfft);
36      fprintf('Input contains a chirp from 50 Hz to 300 Hz and a 150 Hz burst from 0.8 to 1.0 s.\n');
37
```

Figure 1: Experiment 20: STFT and wavelet analysis

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Command Window

```
Sampling frequency = 1000 Hz
STFT window = 128 samples; overlap = 112 samples; FFT length = 512
Input contains a chirp from 50 Hz to 300 Hz and a 150 Hz burst from 0.8 to 1.0 s.
>>
```

Workspace

Name	Value
burst	1x2000 double
F	257x1 double
fCWT	81x1 double
Fs	1000
idx	1x2000 logical
nfft	512
noverlap	112
S	257x118 complex...
t	1x2000 double
Tstft	1x118 double
win	128x1 double
wt	81x2000 complex...
x	1x2000 double

Zoom: 100% UTF-8 CRLF script Ln 36 Col 89

29°C Mostly cloudy

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SASexp18.m | SASexp20.m

```
1   % Experiment 20: STFT and Wavelet-Based Time-Frequency Analysis
2   clc; clear; close all;
3
4   Fs = 1000;
5   t = 0:1/Fs:2-1/Fs;
6
7   % Non-stationary input: linear chirp from 50 Hz to 300 Hz plus a transient
8   x = chirp(t,50,t(end),300,'linear');
9   burst = zeros(size(t));
10  idx = (t >= 0.8) & (t <= 1.0);
11  burst(idx) = 0.8*sin(2*pi*150*t(idx));
12  x = x + burst;
13
14  % STFT / spectrogram
15  win = hann(128,'periodic');
16  noverlap = 112;
17  nfft = 512;
18  [S,F,Tstft] = spectrogram(x,win,noverlap,nfft,Fs);
19  figure('Name','Experiment 20: STFT and wavelet analysis');
20  subplot(3,1,1);
21  plot(t,x); grid on;
22  xlabel('Time (s)'); ylabel('Amplitude');
23  title('Input: Chirp with 150 Hz transient');
24
25  subplot(3,1,2);
26  imagesc(Tstft,F,20*log10(abs(S)+eps)); axis xy;
27  ylim([0 400]); colorbar;
28  xlabel('Time (s)'); ylabel('Frequency (Hz)');
29  title('STFT Magnitude Spectrogram (dB)');
30
```

Command Window | Workspace

Name	Value
burst	1x2000 double
F	257x1 double
Fs	1000
idx	1x2000 logical
nfft	512
noverlap	112
S	257x118 complex...
t	1x2000 double
Tstft	1x118 double
win	128x1 double
x	1x2000 double

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